

Gabriel Burrows

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Education

Queen's University

Bachelor of Computing (Hons.)

Biomedical Computing Specialization

Kingston, Ontario

September 2021 – December 2025

- **Honors:** Dean's Honour List (Recognition for top-tier academic performance)
- **Concentrations:** Software Architecture, Data Science, Biomedical Informatics, Human-Computer Interaction.
- **Related Coursework:** Software Requirements Analysis, Database Management Systems, Linear Algebra, Statistics for Biomedical Data, Computer-Integrated Surgery, Algorithms & Data Structures.

Projects

Deep Brain Stimulus (DBS) Therapy Target Locator Module for 3D Slicer

Kingston, Ontario

Team Lead

September 2025 – December 2025

- **Solution Design:** Engineered a Python-based pipeline for 4D fMRI registration, translating complex clinical requirements into an automated 3D Slicer module.
- **Analytical Thinking:** Identified high-value neural regions in 75% of test cases, improving surgical planning accuracy through advanced image preprocessing and MNI152 template registration.
- **Technical Documentation:** Authored comprehensive module documentation to streamline clinical workflows, enabling non-technical medical staff to identify electrode sites via 3D heatmaps.

Ames Housing Market: Advanced Regression & Predictive Modeling Kaggle Competition

Kingston, Ontario

Team Lead

September 2025 – December 2025

- **Business Intelligence:** Optimized house price predictions using Scikit-Learn pipelines, ranking in the top 44% of 5,400+ participants by engineering domain-specific features.
- **Problem Solving:** Increased model R2 from 0.74 to 0.82 through automated hyperparameter tuning (GridSearchCV) and rigorous outlier analysis to eliminate bias.

Activities and Leadership

Queen's University Japanese 200

Kingston, Ontario

Teaching Assistant

September 2024 – April 2025

- **Communication:** Articulated complex linguistic and grammatical concepts to audiences of varying perception levels, directly supporting the Professor in lesson delivery.
- **Project Management:** Managed grading timelines and student feedback for a cohort of 30+ students, ensuring high-quality academic support and deadline adherence.
- **Bilingual Liaison:** Served as the secondary point of contact for student inquiries, conducting administrative tasks and requirements definition in both English and Japanese.

Japanese Culture Club

Kingston, Ontario

Event Director

September 2021 – April 2024

- **Organization:** Spearheaded the planning and execution of large-scale cultural events (30+ attendees), managing logistics, budgeting, and team coordination.
- **Stakeholder Management:** Collaborated with university administration, external vendors and club finance managers to deliver successful community experiences.

Skills

- **Programming:** Python (Pandas, Scikit-Learn, PyTorch, Django), MATLAB, SQL (MySQL, PostgreSQL), Next.js, Typescript.
- **Software & Tools:** GitHub, Docker, G Suite, MS Office, Lucidchart, Jupyter Notebooks.
- **Technical/Consulting:** System Configuration, Requirements Gathering, Data Analysis, Technical Documentation, Software Design Life Cycle.
- **Languages:** English (Native), Japanese (Business / JLPT N2 - Proficient in written/spoken business contexts).

Interests

- **Life Sciences Tech:** Passionate about the intersection of Software and Healthcare
- **Technology:** Trends in Software Development, AI in medical imaging, Electronics Hardware Enthusiast
- **For Fun:** Hiking, Cooking, Gaming, Photography